

NSG ELECTROSTATS

Instructions

Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.

A. General:

1. This question paper has 30 questions
2. Additional sheets for rough work shall not be provided
3. Blank papers, clipboards, log tables, slide rules, calculators, cellular phones, pagers, and any other electronic gadgets will not be allowed
4. Do not tamper / mutilate the ORS

B. Filling the ORS:

5. Fill in your Roll No., Name and Signature with a pen in appropriate boxes. Do not write these anywhere else
6. Darken the appropriate bubbles below your ID

C. Question paper format and Marking scheme:

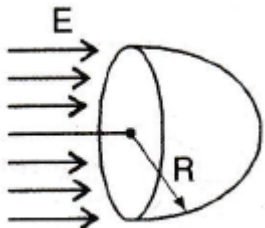
7. The question paper is divided into 1 parts; Physics . Each part consists of two sections.
8. In Section A, each question has four choices (1), (2), (3) and (4) out of which ONLY ONE is correct. For every correct choice marked, 4 marks shall be awarded, and for every wrong choice marked 0 mark will be deducted.
9. In Section B, the answer to each question is a numerical value. For every correct answer, 4 marks shall be awarded, and for every wrong answered 0 mark will be deducted.

Physics

Section A

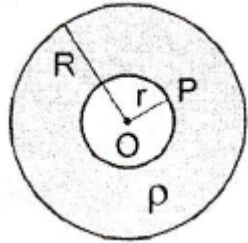
Questions 1-28 contains questions with 4 options (1), (2), (3) and (4), of which only one is correct. Mark the most appropriate choice.

1. According to the figure, a hemispherical object is placed in an electric field, find the outward flux through its curved surface:



2. A cylinder of radius R and length L is placed in a uniform electric field E parallel to the cylinder axis. The total flux for the surface of the cylinder is given by:

3. A insulated sphere of radius R has a uniform volume charge density ρ . The electric field at a point P inside the sphere and distant r from the centre is:

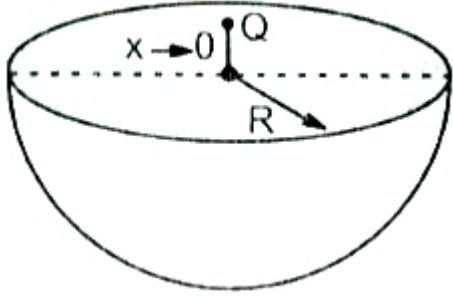


4. If the electric flux entering and leaving an enclosed surface respectively is ϕ_1 and ϕ_2 , the electric charge inside the surface will be_____

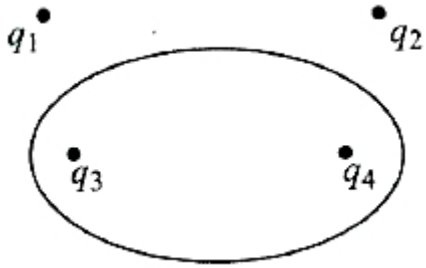
5. A charge Q is situated at the centre of a cube. The electric flux through one of the faces of the cube is_____

6. In a region of space, the electric field is given by $\vec{E} = 8\hat{i} + 4\hat{j} + 3\hat{k}$. The electric flux through a surface of area 100 units in the xy plane is

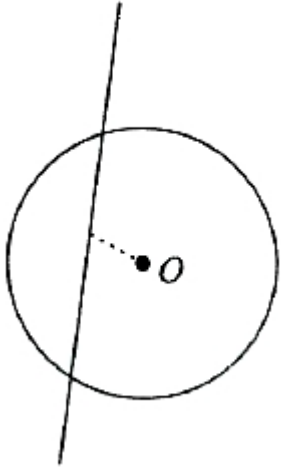
7. A point charge Q is located just above the centre of the flat face of hemisphere as shown in figure. The electric flux through the flat face and curved face of hemisphere are respectively,



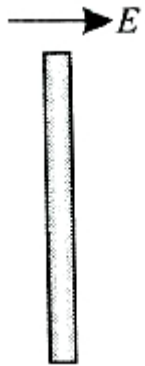
8. Fig. 2.95 shows four charges q_1 , q_2 , q_3 , and q_4 fixed in space. Then the total flux of the electric field through a closed surface S , due to all the charges, is



9. A uniformly charged and infinitely long line having a liner charge density λ is placed at a normal distance y from a point O . Consider a sphere of radius R with O as the center and $R > y$. Electric flux through the surface of the sphere is



10. An uncharged conducting large plate is placed as shown. Now an electric field E toward right is applied. Find the induced charge density on the right surface of the plate.

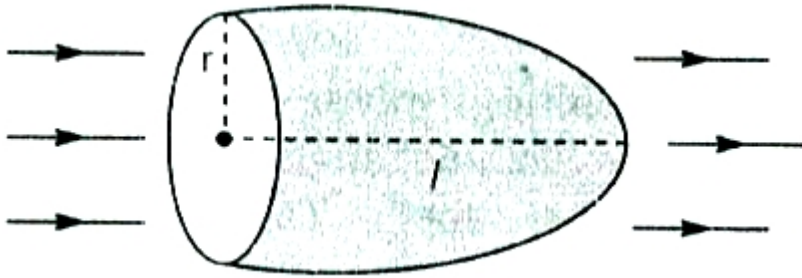


11. At a point in space, the electric field points towards north. In the region surrounding this point, the rate of change of potential will be zero along

12. The electric flux from a cube of edge l is ϕ . If an edge of the cube is made $2l$ and the charge enclosed is halved, its value will be

13. A metallic shell has a point charge q kept inside its cavity. Which of the following diagrams correctly represents the electric lines of forces?

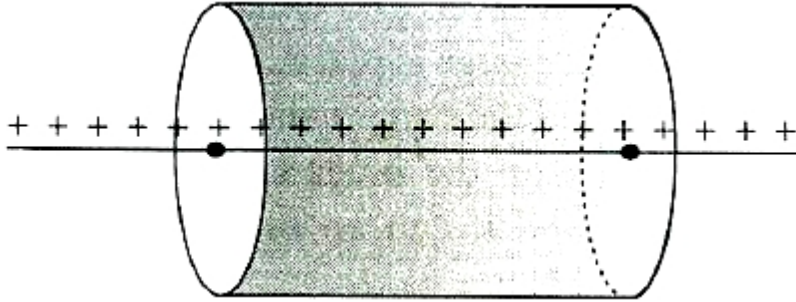
14. Fig. shows a circular surface and a paraboloidal surface. It is placed in a uniform electric field of magnitude E such that the circular surface is oriented at right-angles to the direction of field. Electric flux through the paraboloidal surface is:



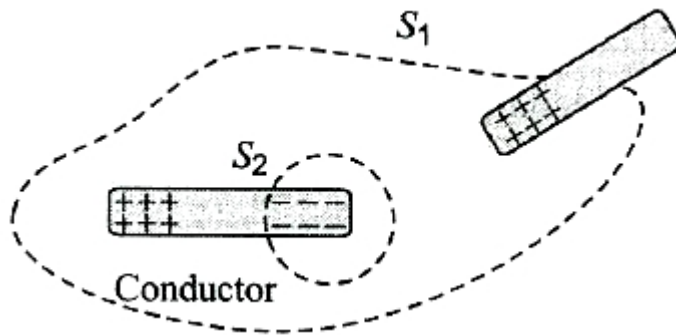
15. A cylinder of radius R and length l is placed in a uniform electric field E parallel to the axis of the cylinder. The total flux over the curved surface of the cylinder is

16. A system consists of a thin charged wire ring of radius r and a very long uniformly charged wire oriented along the axis of the ring, with one of its ends coinciding with the center of the ring. The total charge on the ring is q , and the linear charge density on the straight wire is λ . The interaction force between the ring and the wire is

17. Consider an infinite line charge having uniform linear charge density and passing through the axis of a cylinder. What will be the effect on the flux passing through the curved surface if the portions of the line charge outside the cylinder is removed.



18. Charge on an originally uncharged conductor is separated by holding a positively charged rod very closely nearby, as shown in

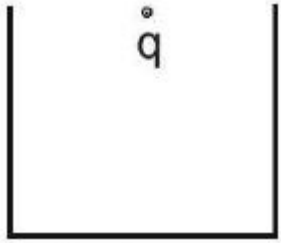


. Assume that the induced negative charge on the conductor is equal to the positive charge q on the rod. Then the flux through surface S_1 is

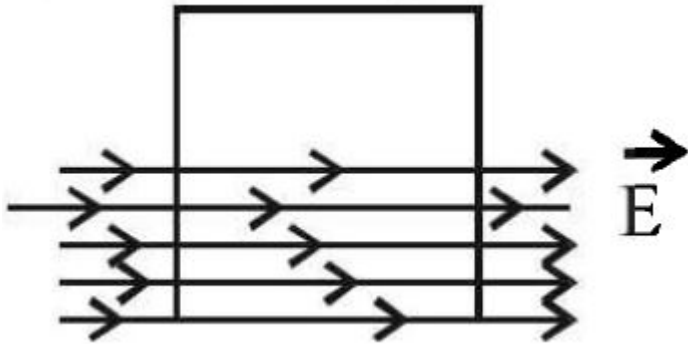
19. A cube of side 10 cm encloses a charge of $0.1\mu\text{C}$ at its center. Calculate the number of lines of force through each face of the cube.

20. Consider two concentric spherical surfaces S_1 with radius a and S_2 with radius $2a$, both centered at the origin. There is a charge $+q$ at the origin and there are no other charges. Compare the flux ϕ_1 through S_1 with the flux ϕ_2 through S_2 .

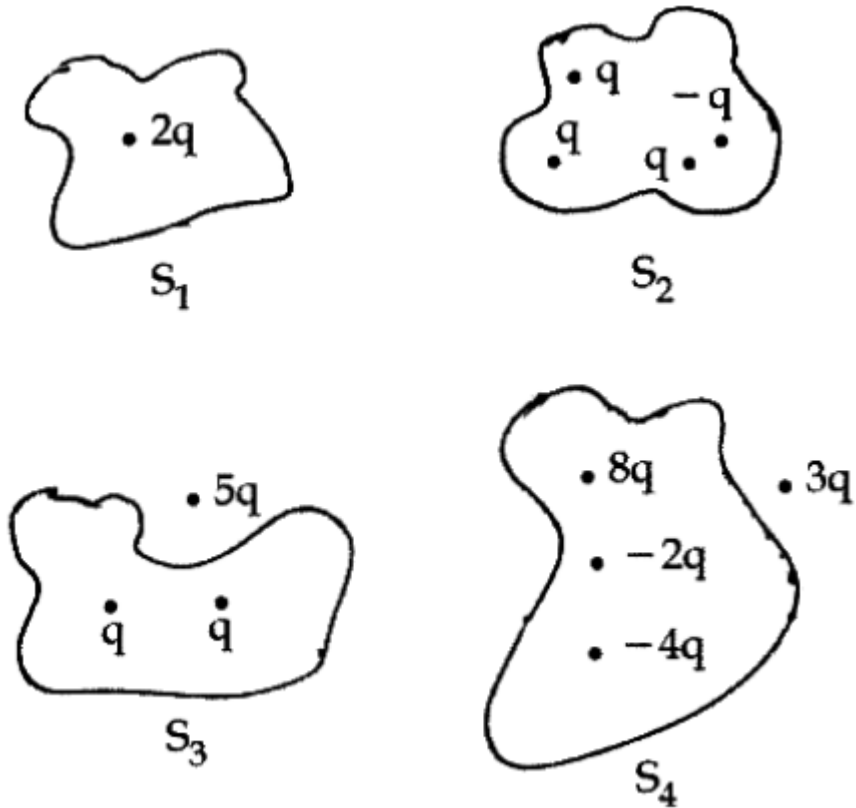
21. A charge q is placed at the centre of the open end of a cylindrical vessel as shown in fig. The flux of the electric field through the surface of the vessel is



22. A square surface of side L meters is in the plane of the paper. A uniform electric field $\vec{E}$ (volt/m), also in the plane of the paper, is limited only to the lower half of the square surface (see figure). The electric flux in SI units associated with the surface is

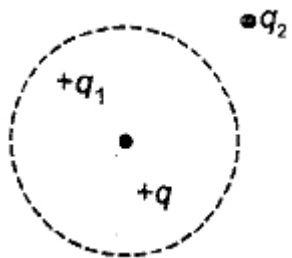


23. Four closed surfaces and corresponding charge distributions are shown below -

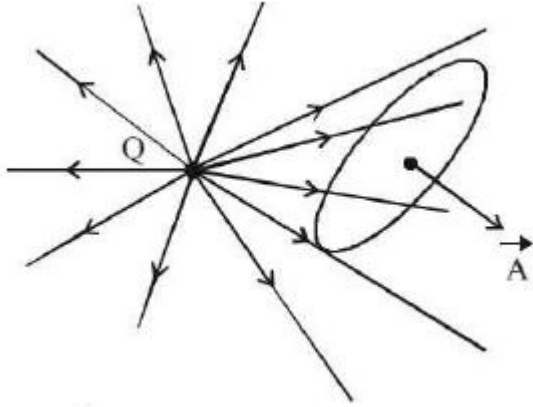


Let the respective electric fluxes through the surfaces be Φ_1 , Φ_2 , Φ_3 and Φ_4 Then -

24. Consider the charge configuration and a spherical gaussian surface as shown in the figure. When calculating the flux of the electric field over the spherical surface, the electric field will be due to



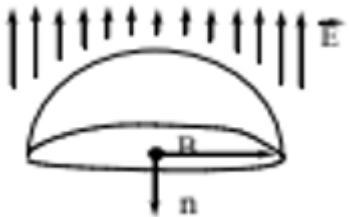
25. In the figure, the net electric flux through the area A is $\phi = \vec{E} \cdot \vec{A}$ when the system is in air. On immersing the system in water the net electric flux through the area



26. The electric potential in a certain region of space depends only on x coordinate as $V = -\alpha x^3 + \beta$, where α and β are constants. The volume charge distribution of space charge is given as:

27. A negative charge is placed at the centre of the non-conducting sphere. The direction of the electric field on any point at the surface of the sphere is

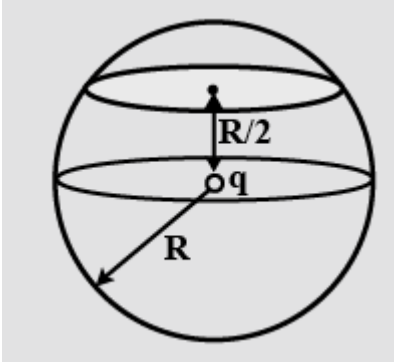
28. If a hemispherical body is placed in a uniform electric field E then the flux linked with the curved surface is



Section B

Questions 29–30 contains questions of which, the answer would be a numerical value.

29. Flux passing through the shaded surface of a sphere is $q/Z\epsilon_0$ when a point charge q is placed at the center. (radius of the sphere is R) Find the value of Z



30. One-fourth of a sphere of radius $1m$ is removed as shown in the figure. An electric field $\frac{\sqrt{2}}{\pi}$ N/C exists parallel to the xy plane. Find the flux through the remaining curved part is (in Nm^2/C)

